

Experiment 1

Kali Linux Installation (VMware / VirtualBox)

AIM

To install Kali Linux operating system on a virtual machine using VMware Workstation or VirtualBox and verify the installation by running basic commands.

REQUIREMENTS

Software	VMware Workstation 17 / Oracle VirtualBox 7.0
ISO Image	kali-linux-2024.1-installer-amd64.iso (Official Download)
RAM	Minimum 4 GB (8 GB Recommended)
Disk Space	Minimum 20 GB Free Space
Processor	Intel i5 or above / AMD Ryzen 5 or above
Network	NAT / Bridged Adapter

THEORY

Kali Linux is a Debian-based Linux distribution specifically designed for digital forensics and penetration testing, developed and maintained by Offensive Security. It contains over 600 pre-installed tools including Nmap, Wireshark, Metasploit, Aircrack-ng, Burp Suite, and John the Ripper.

A Virtual Machine (VM) is a software-based computer that runs on top of a physical host. Using VMware or VirtualBox, Kali Linux can be installed inside Windows without affecting the main OS — ideal for practising ethical hacking safely. Key benefits include an isolated environment, easy snapshot and rollback, no need to dual-boot, and portable VM images.

PROCEDURE

- Download VMware Workstation from [vmware.com](https://www.vmware.com) or VirtualBox from [virtualbox.org](https://www.virtualbox.org).
- Download the Kali Linux ISO image from <https://www.kali.org/get-kali/>
- Open VMware ® Click 'Create a New Virtual Machine' ® Choose Typical.
- Select the downloaded Kali ISO as the installer disc image.
- Set Guest OS as 'Linux' ® Version: Debian 10.x 64-bit.
- Set VM Name as 'Kali-Linux-2024' and choose an installation location.
- Allocate disk space – minimum 20 GB. Select 'Store as single file'.
- Customise hardware: RAM = 4096 MB, Processors = 2, Network = NAT.
- Power on the VM; the Kali installer will boot from the ISO.
- Choose 'Graphical Install' from the boot menu.
- Select Language: English, Country: India, Keyboard: American English.
- Set hostname as 'kali'; domain name may be left blank.
- Create root user and set a strong password.

- Partition disks ® Use entire disk ® All files in one partition ® Finish.
- Wait for base system installation (approximately 10–15 minutes).
- Select software to install: Desktop environment (GNOME/XFCE) + Standard tools.
- Install GRUB bootloader ® Select /dev/sda.
- Click Continue after 'Installation complete' to reboot.
- Login with username: root and the password set during installation.
- Open Terminal and run the verification commands below.

COMMANDS

```
# Check Kali Linux version
cat /etc/os-release

# Check kernel version
uname -r

# Check IP address
ifconfig

# Check network connectivity
ping -c 4 google.com

# List installed security tools
ls /usr/bin | grep -E 'nmap|metasploit|john|aircrack'

# Update the system
apt update && apt upgrade -y
```

OUTPUT

Screenshot: Terminal – System Verification Commands

```
root@kali:~# cat /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION="2024.1"

root@kali:~# uname -r
6.6.9-amd64

root@kali:~# ifconfig
eth0: flags=4163 mtu 1500
inet 192.168.1.105 netmask 255.255.255.0

root@kali:~# ping -c 4 google.com
64 bytes from 142.250.77.78: icmp_seq=1 ttl=115 time=14.3 ms
64 bytes from 142.250.77.78: icmp_seq=2 ttl=115 time=13.8 ms
4 packets transmitted, 4 received, 0% packet loss

root@kali:~# apt update
Hit:1 http://kali.download/kali kali-rolling InRelease
All packages are up to date.
```

RESULT

Kali Linux 2024.1 was successfully installed on VMware Workstation. The system booted correctly, network connectivity was confirmed with ping, and the OS version was verified using `cat /etc/os-release`. The lab environment is ready for further security experiments.